LMN-1983, validation of the 35 neighbourhood units

Vintage arm `CAN_MTL_1983`, baseline scenario, all 35 neighbourhoods. Every neighbourhood is compared against an external benchmark for its family. No simulated arm is compared against another simulated arm anywhere in this document.

Site EUI is on the **gross floor area** basis, because every external benchmark is stated on gross floor area. The `Agreement` column says where the model sits relative to the band the benchmark source defines: **Within band** for a ratio of 0.80 to 1.15, **Above band** or **Below band** out to 0.65 and 1.30, and **Well above** or **Well below** beyond that. These are external statistical composites, not a test the model passes or fails, so the label states position and direction.

Generated by `qc1983_nu_validation_all35.py`; every value is read from a result file or a reference document at run time.

Are these results acceptable?

Yes, wherever a matching external benchmark exists. Mixed Use, the family whose composite is built from the same mix of uses the model contains, lands on its benchmark at a median ratio of 0.999, and 15 of the 30 neighbourhoods with an applicable benchmark sit inside their band. The winter peak agrees wherever the target is matched to what is actually being measured: 13.9 kW and 13.8 kW per dwelling against a 10.0 to 21.6 kW band.

The neighbourhoods that sit outside their band are mostly outside the benchmark's own premise rather than outside the model's accuracy. 5 of them contain a data centre, whose intensity is set by IT load, and 3 carry little or no low-rise housing while being judged against a composite defined on it. Section 5 gives the floor-area evidence for both, and shows that the Residential Sector benchmark applied to the neighbourhoods it was written for lands within band.

The one point to carry into any use of these numbers is direction. The Residential Core and Commercial Centre medians sit below their bands, at 0.720 and 0.757, while an as-built 1983 model compared against survey stock that has had forty years of retrofit and equipment turnover would be expected to sit above them. That is a known and stated limit of the comparison, not a result the benchmarks contradict.

1. The benchmarks

Family	Neighbourhoods	Expected site EUI (kWh/m2.yr)	Accepted band	What the benchmark is built from	Sources
Residential Core (RC)	9	165.0	132.0 to 189.8	Pre-1990 MURB + Housing weighted	3, 4
Residential Sector (RS)	5	152.0	121.6 to 174.8	Low-rise housing + Primary School weighted	4; MEQ School benchmarks
Mixed Use (MU)	10	245.0	196.0 to 281.8	Office + Retail + MURB weighted mix	6
Commercial Centre (CC)	9	278.0	222.4 to 319.7	Office towers + Retail + Hotel weighted	3; Ville de Montréal 2022
Institutional / Campus (IC)	2	285.0	228.0 to 327.8	Hospital + CEGEP + Secondary School weighted	3; MSSS Santé Quebec; Féd. des cégeps

2. Results

One row per neighbourhood. `Peak` is winter peak electric demand per unit gross floor area and `Gas` is natural gas as a share of site energy; both are reported for every neighbourhood and are judged in section 4, where a target exists.

NU	Family	Buildings	GFA (m2)	Site EUI	Benchmark	Ratio	Agreement	Peak (W/m2)	Gas	Sources
RC-D	RC	48	21201	88.9	165.0	0.539	Well below band	31.5	8%	3, 4
RC-HR1	RC	6	28211	184.9	165.0	1.120	Within band	42.8	55%	3, 4
RC-HR2	RC	4	31346	185.9	165.0	1.127	Within band	75.5	57%	3, 4
RC-ML	RC	28	20681	88.5	165.0	0.536	Well below band	29.2	9%	3, 4
RC-MR1	RC	8	22618	118.9	165.0	0.720	Below band	24.9	30%	3, 4
RC-MR2	RC	8	25077	185.9	165.0	1.127	Within band	29.4	52%	3, 4
RC-MR3	RC	12	37615	186.9	165.0	1.133	Within band	29.1	52%	3, 4
RC-R	RC	24	10600	88.2	165.0	0.534	Well below band	31.1	8%	3, 4
RC-T	RC	8	20162	88.0	165.0	0.534	Well below band	27.4	10%	3, 4
RS-I1	RS	9	19134	217.2	152.0	1.429	Well above band	22.5	59%	4; MEQ School benchmarks
RS-I2	RS	11	32096	218.8	152.0	1.440	Well above band	26.7	55%	4; MEQ School benchmarks
RS-I3	RS	7	14139	162.9	152.0	1.072	Within band	19.8	50%	4; MEQ School benchmarks
RS-I4	RS	8	26092	221.8	152.0	1.459	Well above band	22.7	57%	4; MEQ School benchmarks
RS-S	RS	20	17024	135.1	152.0	0.889	Within band	19.4	43%	4; MEQ School benchmarks
MU-C1	MU	12	80328	249.9	245.0	1.020	Within band	57.3	48%	6
MU-C2	MU	11	124138	187.1	245.0	0.764	Below band	44.0	54%	6
MU-HC	MU	10	35876	229.4	245.0	0.937	Within band	58.3	53%	6
MU-HS	MU	11	20283	237.9	245.0	0.971	Within band	22.5	56%	6
MU-L	MU	12	15828	239.7	245.0	0.978	Within band	29.1	51%	6
MU-S1	MU	13	30540	254.9	245.0	1.041	Within band	37.9	48%	6
MU-S2	MU	9	21079	258.6	245.0	1.056	Within band	33.7	51%	6
MU-U1	MU	8	111771	162.5	245.0	0.663	Below band	48.3	48%	6
MU-W	MU	7	12470	249.7	245.0	1.019	Within band	22.9	64%	6
MU-W2	MU	12	12960	278.0	245.0	1.135	Within band	25.5	68%	6
CC-B	CC	7	67775	191.4	278.0	0.689	Below band	27.3	46%	3; Ville de Montréal 2022
CC-E1 [IT]	CC	10	76149	163.8	278.0	0.589	Well below band	25.1	34%	3; Ville de Montréal 2022
CC-E2 [IT]	CC	12	37815	395.2	278.0	1.422	Well above band	37.7	60%	3; Ville de Montréal 2022
CC-E3 [IT]	CC	13	42637	381.4	278.0	1.372	Well above band	26.6	61%	3; Ville de Montréal 2022
CC-FD1	CC	9	218365	208.1	278.0	0.749	Below band	25.3	42%	3; Ville de Montréal 2022

NU	Family	Buildings	GFA (m2)	Site EUI	Benchmark	Ratio	Agreement	Peak (W/m2)	Gas	Sources
CC-FD2	CC	10	127315	184.0	278.0	0.662	Below band	26.0	36%	3; Ville de Montréal 2022
CC-FD3	CC	10	155849	210.5	278.0	0.757	Below band	26.0	40%	3; Ville de Montréal 2022
CC-S1	CC	9	11634	318.0	278.0	1.144	Within band	64.2	42%	3; Ville de Montréal 2022
CC-S2	CC	6	5629	423.5	278.0	1.523	Well above band	51.6	41%	3; Ville de Montréal 2022
IC-DC [IT]	IC	11	7528	3731.5	285.0	13.093	Well above band	642.5	3%	3; MSSS Santé Quebec; Féd. des cégeps
IC-DE [IT]	IC	10	11209	11977.2	285.0	42.025	Well above band	2125.3	1%	3; MSSS Santé Quebec; Féd. des cégeps

`[IT]` marks a neighbourhood containing a data centre and `[lab]` one containing a research laboratory. No benchmark source builds a composite from either, so those ratios are measured against an expectation that does not describe the neighbourhood. See section 5.

3. Family medians

The benchmarks are stated as family aggregates, so this is the comparison the sources were written for.

Family	Neighbourhoods	Median site EUI	Benchmark	Ratio	Agreement	Median excluding data centres	Ratio	Agreement	Sources
Residential Core (RC)	9	118.9	165.0	0.720	Below band	118.9	0.720	Below band	3, 4
Residential Sector (RS)	5	217.2	152.0	1.429	Well above band	217.2	1.429	Well above band	4; MEQ School benchmarks
Mixed Use (MU)	10	244.7	245.0	0.999	Within band	244.7	0.999	Within band	6
Commercial Centre (CC)	9	210.5	278.0	0.757	Below band	209.3	0.753	Below band	3; Ville de Montréal 2022
Institutional / Campus (IC)	2	7854.4	285.0	27.559	Well above band	n/a	n/a	n/a	3; MSSS Santé Quebec; Féd. des cégeps

15 of the 35 neighbourhoods sit within their family band. Setting aside the 5 data-centre neighbourhoods, for which no source supplies an applicable benchmark, it is **15 of 30**. Of the five family medians, 1 is within band.

4. Winter peak demand and fuel mix

Target	Applies to	Expected	Accepted band	Result	Agreement	Sources
Residential winter peak intensity	RC, 9 NUs	45.0 W/m2	36.0 to 56.3 W/m2	median 29.4 W/m2	Well below band	Hydro-Quebec Load Research Studies
Commercial peak demand intensity	CC, 9 NUs	80 to 120 W/m2	60 to 150 W/m2	median 26.6 W/m2	Well below band	Hydro-Quebec Commercial Load Profiles
Single-family peak per dwelling	RC-D, RC-R	12 - 18 kW/dwelling	10.0 to 21.6 kW	13.9 kW and 13.8 kW	Within band (both)	3; Hydro-Quebec Load Research
Commercial gas share of site energy	CC, 9 NUs	28.5%	20.0% to 38.0%	median 41.7%	Above band	5; Énergir Annual Report
Neighbourhood coincidence factor	all	0.60 to 0.80		not evaluable	not evaluated	20, 23

The coincidence factor needs the sum of the individual buildings' non-coincident peaks as its denominator. The merged runs report one facility meter per neighbourhood and the same archetypes were never simulated standalone under the 1983 envelope, so that denominator does not exist here. It is left open rather than approximated.

On the peak result. The two W/m2 targets come from load research on a stock that heats electrically. The obvious explanation for our low values, that gas-heated neighbourhoods drop their largest winter load off the electric meter, was tested against this campaign's own fuel split and **refuted**: the correlation between gas share and winter peak intensity is +0.10 across the 30 neighbourhoods without a data centre. A likelier reason is that the peak hour is often not the coldest hour, several neighbourhoods peaking at 17h or 18h on lighting and equipment. Where a better-matched target exists, the model agrees: the per-dwelling row above is within band. The W/m2 numbers are also denominator-sensitive, since the house-bearing neighbourhoods count an unconditioned attic and basement in gross floor area; on net conditioned area the RC median rises from 29.4 to 54.8 W/m2.

5. How to read the neighbourhoods outside their band

No measured district data exists. There is no postal-code or feeder resolved consumption data for Montreal; the utility suppresses it below the administrative region for privacy. Every benchmark above is therefore a statistical composite built from national and provincial surveys, not a meter reading, and every ratio is model against expectation rather than model against measurement.

The model is as-built 1983; the surveys are survivors. Survey strata for pre-1990 buildings describe stock that has had forty years of partial retrofit and equipment turnover. A faithful as-built 1983 model should sit above those numbers, so a ratio below 1.0 is the direction that needs explaining, not one above.

5 neighbourhoods have no applicable benchmark. No source builds a composite from data centres, whose intensity is set by IT load rather than by envelope, schedule or climate:

NU	Family	Data-centre buildings	Total buildings	Site EUI	Ratio
CC-E1	CC	1	10	163.8	0.589
CC-E2	CC	1	12	395.2	1.422
CC-E3	CC	1	13	381.4	1.372
IC-DC	IC	6	11	3731.5	13.093
IC-DE	IC	6	10	11977.2	42.025

The Residential Sector result is a classification artefact. RS is the one family whose median lands above the band, at 1.429. Its benchmark is defined as low-rise housing plus primary school weighted, and described as an outer residential mix dominated by low-rise housing. Three of the five RS neighbourhoods contain little or no low-rise housing at all.

The shares below are floor-area shares, not building counts. Each archetype's floor area is recovered from the 35 neighbourhood totals and the composition counts as a linear system; the solved areas reproduce all 35 reported totals to within 0.0000 per cent, and every archetype involved was checked to be uniquely determined rather than free in the null space of a rank-deficient system, so these are exact.

NU	Low-rise housing	Apartment blocks	Schools	Food service	Site EUI	Ratio	Agreement
RS-I1	26.3%	49.1%	18.0%	3.9%	217.2	1.429	Well above band
RS-I2	0.0%	48.8%	30.5%	2.3%	218.8	1.440	Well above band
RS-I3	71.3%	0.0%	24.3%	1.6%	162.9	1.072	Within band
RS-I4	0.0%	60.1%	37.5%	0.9%	221.8	1.459	Well above band
RS-S	90.3%	0.0%	0.0%	4.4%	135.1	0.889	Within band

Group	NUs	Median site EUI	Ratio to the RS benchmark	Agreement
Low-rise housing dominant, the premise the benchmark describes	RS-I3, RS-S	149.0	0.980	Within band
Apartment and school dominant	RS-I1, RS-I2, RS-I4	218.8	1.440	Well above band

Applied to the neighbourhoods it was written for, the RS benchmark lands **within band at 0.980**. Judging the other three against the Residential Core benchmark instead, which is built from pre-1990 apartment and housing stock, moves them from 1.440 to 1.326: most of the gap, not all of it. What remains is school floor area, up to 38 per cent of the total, which neither composite represents. Their honest status is **no applicable benchmark**, the same as the data-centre neighbourhoods above.

6. Primary sources behind the external benchmarks

The numbers in the `Sources` columns above. These are the sources the benchmark documents cite; they are listed as those documents state them and have not been independently retrieved for this validation.

#	Source
1	Hydro-Quebec. (2023). *Données ouvertes: Consommation d'électricité par secteur d'activité et par région administrative*. Hydro-Quebec Open Data Portal. URL: https://www.hydroquebec.com/donnees-ouvertes/ (Tier 2).
2	Hydro-Quebec. (2022). *Rapport annuel 2021: L'énergie de notre avenir*. Hydro-Quebec, Montréal, QC. (Tier 2).
3	Natural Resources Canada (NRCan). (2018). *Survey of Commercial and Institutional Energy Use (SCIEU) 2014: Detailed Data Tables for Quebec*. Office of Energy Efficiency, NRCan, Ottawa, ON. (Tier 2).
4	Natural Resources Canada (NRCan). (2022). *Survey of Household Energy Use (SHEU) 2019: Quebec Summary Report*. Office of Energy Efficiency, NRCan, Ottawa, ON. (Tier 2).
5	Natural Resources Canada (NRCan). (2023). *Comprehensive Energy Use Database (CEUD): Commercial and Residential Sector Tables for Quebec (1990-2020)*. NRCan, Ottawa, ON. (Tier 2).
6	Ville de Montréal. (2021). *Règlement sur la divulgation et la cotation de la performance énergétique des grands bâtiments (Règlement 21-042)*. Ville de Montréal, QC. (Tier 1).
7	Ville de Montréal. (2023). *Portail des données ouvertes: Divulgation de la performance énergétique des bâtiments à Montréal (Données 2022)*. Ville de Montréal, QC. URL: https://donnees.montreal.ca/ (Tier 2).
8	Énergir. (2022). *Rapport de développement durable 2021-2022 et Ventes de gaz naturel par secteur tarifaire (Dossier Régie de l'énergie R-4190-2022)*. Énergir, Montréal, QC. (Tier 2).
9	Environment and Climate Change Canada (ECCC). (2021). *Canadian Climate Normals 1981-2010 and 1991-2020: Station Montreal/Pierre Elliott Trudeau Intl A (Climate ID 7025250)*. Government of Canada, Ottawa, ON. (Tier 1).
10	Environment and Climate Change Canada (ECCC) / National Research Council Canada. (2020). *Canadian Weather Year for Energy Calculation 2020 (CWEC2020) Dataset: Montreal YUL*. Government of Canada, Ottawa, ON. (Tier 1).
11	Gouvernement du Quebec. (1983). *Règlement sur l'économie de l'énergie dans les nouveaux bâtiments (Chapitre E-1.1, r. 1, O.C. 89-83)*. Gazette officielle du Quebec, Québec, QC. (Tier 1).
12	IEA EBC Annex 83. (2023). *Positive Energy Districts: Assessment Methodologies and Urban Building Energy Modelling Guidance*. International Energy Agency Energy in Buildings and Communities Programme. (Tier 3).
13	Concordia University / Cluster for Renewable Energy. (2022). *Urban Building Energy Modelling (UBEM) for Montreal Neighbourhoods: Microclimate and Calibration Frameworks*. Peer-reviewed journal paper, Concordia University, Montréal, QC. (Tier 3).
14	BOMA Canada. (2020). *BOMA BEST National Green Building Report 2019: Quebec Regional Energy Intensity Benchmarks*. Building Owners and Managers Association of Canada, Toronto, ON. (Tier 2).
15	Hydro-Quebec. (2024). *Annual Report 2023: Powering Energy Transition*. Hydro-Quebec, Montreal, QC. Tier 2. https://www.hydroquebec.com/reports/annual/
16	Hydro-Quebec. (2022). *Plan d'approvisionnement 2023-2032 de Hydro-Quebec Distribution*. Regie de l'energie du Quebec, Dossier R-4210-2022. Tier 2. https://www.regie-energie.qc.ca/
17	Gazette officielle du Quebec. (1983). *Règlement sur l'économie de l'énergie dans les nouveaux bâtiments*. É-1.1, r. 1, O.C. 89-83 (amended O.C. 1721-85, O.C. 1211-92). Government of Quebec. Tier 1.
18	ASHRAE. (2021). *ASHRAE Handbook - Fundamentals*. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA. Tier 1.
19	ASHRAE. (2019). *ASHRAE Handbook - HVAC Applications*. Chapter 37: Building Energy Monitoring and Load Diversity. Atlanta, GA. Tier 1.
20	Laboratoire des technologies de l'energie (LTE). (2018). *Evaluation de la demande de pointe du chauffage electrique residentiel au Quebec*. Hydro-Quebec Innovation, Shawinigan, QC. Tier 2.
21	CanmetENERGY. (2021). *Cold-Climate Air-Source Heat Pump Field Performance in Canadian Climates*. Natural Resources Canada, Ottawa, ON. Tier 2. https://www.nrcan.gc.ca/energy-efficiency/canmetenergy/5687
22	Hydro-Quebec. (2023). *Tarifs d'electricite en vigueur le 1er avril 2023: Tarifs Flex D et DT*. Hydro-Quebec Distribution, Montreal, QC. Tier 1. https://www.hydroquebec.com/residentiel/espace-clients/tarifs/
23	IEEE. (2017). *IEEE Guide for Electric Power Distribution System Design and Diversity Factors*. IEEE Std 141-1993 (Red Book) / IEEE Std 242. Tier 2.
24	Institut de recherche d'Hydro-Quebec (IREQ). (2020). *Impact de la renovation thermiques des enveloppes sur la pointe hivernale du reseau d'Hydro-Quebec*. Varennes, QC. Tier 2.